MIL-R-13028B(CA)

9 October 1980
Superseding
MIL-R-13028A(SigC)
5 December 1959

MILITARY SPECIFICATION

RADIO SET CONTROL GROUP

(CONTROL GROUP AN/GRA-6( ))
(COMPONENTS FOR)

This specification is approved for use by Communications Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the following components of Control Group AN/GRA-6( ):

Remote Control	C-433( )/GRC
Local Control	C-434( )/GRC
Interconnecting Box	J-654( )/GRC
Bag	CW-189( )/GR

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-152	Treatment, Moisture and Fungus-resistant
MIL-S-901	Shockproof Equipment, Class HI (High-Impact), Shipboard Application, Tests For.
MIL-P-11268	Parts, Materials, And Processes Used in Electronic Equipment
MIL-G-11994	Generator, Ringing, Hand G-42( )/PT
MIL-P-12292	Packaging of Control Group AN/GRA-6
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Electronics Command, ATTN: DRDCO-CM-DM, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5820

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:58Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

DRAWINGS

US Army Electronics Command

SC-DL-40635	Bag CW-189( )/GR
SC-DL-41747	Control Group AN/GRA-6( )
SC-DL-42368	Local Control C-434( )/GRC
SC-DL-42454	Remote Control C-433( )/GRC
SC-DL-85578	Interconnecting Box J-654( )/G

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction. The units covered by this specification shall be constructed in accordance with the applicable drawings listed below:

SC-DL-40635	Bag CW-189( )/GR
SC-DL-41747	Control Group AN/GRA-6( )
SC-DL-42368	Local Control C-434( )/GRC
SC-DL-42454	Remote Control C-433( )/GRC
SC-DL-85578	Interconnecting Box J-654( )/G

3.3 Tropicalization.

3.3.1 Treatment of assembled equipment. Assembled equipment shall be treated in accordance with Specification MIL-T-152 unless both of the following conditions are met:

a. Parts and materials have been tropicalized in accordance with MIL-P-11268.

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:58Z
Check the source to verify that this is the current version before use.

b. The finish of the interior equipment meets the requirements of Specification MIL-F-14072.

3.4 Finish. Units shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings (See 4.4).

3.5 Marking.

3.5.1 General. Marking shall conform to Specification MIL-M-13231 (See 4.4).

3.5.2 Visibility. Whenever practicable, parts and assemblies shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial numbers. The following articles shall be serial numbered: Control C-433( )/GRC and Control C-434( )/GRC.

3.6 Air seal. When tested in accordance with 4.6, the decrease in vacuum shall not exceed 0.01 pounds per square inch, gauge (PSIG), per minute (See 4.6).

3.7 Dielectric strength. The equipment shall withstand 400 volts or two times the maximum peak operating voltage, whichever is greater, for at least ten seconds (See 4.15).

3.8 Operational inspection. The equipment shall meet the operational requirements of 4.16.

3.9 Interconnection box J-654( )/GRC. The box shall be wired in accordance with drawing SC-C-85579 (See 4.17).

3.10 Bag CW-189( )/GR. The bag shall provide spaces for the following equipments in accordance with SC-DL-40635 (See 4.18):

1 each Interconnecting Box J-654( )/G
1 each Control C-433( )/GRC
1 each Control C-434( )/GRC
1 each Handset H-33( )/PT

3.11 Generator, Ringing, Hand G-42( )/PT. In addition to meeting all requirements of this specification, the generator shall meet the following requirements of Specification MIL-G-11994 (See 4.19):

3.5.1.1 Insulation
3.5.1.2 Open circuit voltage
3.5.1.3 Power output
3.5.1.4 Overspeed
3.5.1.5 Harmonic content
3.5.2.1 Operations
3.5.2.2 Frequency
3.5.2.3 Starting torque
3.5.2.4 Switch operation
3.5.2.5 Switch click operation

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:58Z
Check the source to verify that this is the current version before use.

3.12 Service conditions. The equipment shall meet the following service conditions (where a test is referenced, meeting the test shall be considered as compliance with this requirement):

3.12.1 Temperature.

3.12.1.1 Low. During the test of 4.7, the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
3.8	Operational inspection	Full specification performance
3.11	Generator, Ringing, Hand G-42( )/PT (Open circuit voltage and Switch click elimination only)	Full specification performance

After testing, the equipment shall meet full specification performance for these measurements.

3.12.1.2 High. During the test of 4.8, the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
3.8	Operational inspection	Full specification performance
3.11	Generator, Ringing, Hand G-42( )/PT (Open circuit voltage and Switch click elimination only)	Full specification performance

After testing, the equipment shall meet full specification performance for these measurements.

3.12.2 Humidity. During the test of 4.9, the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
3.8	Operational inspection	Full specification performance
3.11	Generator, Ringing, Hand G-42( )/PT (Insulation only)	Full specification performance

Before and after testing, the equipment shall meet full specification performance for the Insulation, Open circuit voltage, Power output and Overspeed requirements of 3.11.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:58Z
Check the source to verify that this is the current version before use.

3.12.3 Elevation. The equipment shall meet full specification performance for the following measurements both during and after testing as specified in 4.10:

Paragraph	Measurement
3.8	Operational inspection
3.11	Generator, Ringing, Hand G-42( )/PT (Open circuit voltage only)

3.12.4 Leakage (immersion). The equipment shall show no evidence of leakage after it is tested as specified in 4.11.

3.12.5 Vibration, bounce and shock. The equipment shall meet the requirements of Table I.

TABLE I - Bounce and shock

Paragraph heading	Test para	Performance after test
Bounce (vibration)	4.12	Specified performance (Notes A & B) No physical damage
Shock, ballistic	4.13	Specified performance (Note C) Minor mechanical damage is permissible.
Shock; bench-handling	4.14	Specified performance (Note A) No physical damage

NOTE A: With the equipment shock-mounts (if any) blocked, the amplitude of vibration of any part, subassembly or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 hertz.

NOTE B: The equipment shall meet full specification performance for 3.8 Operational inspection after testing as specified in 4.12 or 4.14.

NOTE C: After testing as specified in 4.13, the local and remote control components shall be operated as field telephone units. Satisfactory communication shall be maintained before and after testing as specified in 4.13.

3.13 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454 (See 4.20).

3.14 Preconditioning. The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.21 (Also see 4.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:58Z
Check the source to verify that this is the current version before use.

3.15 Workmanship. The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

a. In MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
Slack
Protection
Clearance
Splicing
Connections, general
Grounding, general
Shielding on wire and cable
Soldering and brazing
Cleaning
General requirements for securing of parts
Cleaning prior to welding
Process for welding
Cleaning after welding
Controls
Self-tapping screws

b. In this specification:

3.2 Construction
3.3 Tropicalization
3.4 Finish
3.5 Marking

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (See 4.3). Does not include packaging.

b. Inspection covered by subsidiary documents (See 4.4).

c. Quality conformance inspections:

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:58Z
Check the source to verify that this is the current version before use.

(1) Quality conformance inspection of equipment before packaging (See 4.5).

(2) Quality conformance inspection of packaging (See 4.23).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish two (2) first article units of the AN/GRA-6.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, preconditioning listed in 4.21, and the inspections specified for Group A, Group B and Group C (See Tables III, IV and V respectively). The inspection shall be performed in the following order: 4.4, 4.21, Group A, Group B and Group C as specified in Table II.

TABLE II - Order of first article Group C inspection

Inspection	Test Sequence
Low temperature	3
High temperature	2
Humidity	4
Elevation	1
Bounce	5
Shock; ballistic	7
Shock; bench-handling	6

4.3.3 First article data. The first article test plan and test report shall be as required in the contract or purchase order.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirement specified in the contract or purchase order.

Item	Where required
Finish	3.4
Marking	3.5

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:58Z
Check the source to verify that this is the current version before use.

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, 4.21 and 4.5.1 through 4.5.3. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table III. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subjected to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table III.

TABLE III - Group A inspection

Inspection	Req Para	Test Para	AQL Major	AQL Minor
Visual and mechanical	3.15	4.22		
Remote control C-433/GRC			1.0%	4.0%
Local control C-434/GRC			1.5%	6.5%
Interconnection Box J-654/G			1.0%	4.0%
Bag CW-189/GR			1.0%	4.0%
Fit of components in CW-189/GR	3.10	4.18	1.0%	----
Generator, Ringing, G-42( )/PT	3.11	4.19		
Click elimination			1.0% for the group	4.0% for the group
Starting torque				
Switch operation				
Air Seal (See Note)			1.0%	----
Electrical				
Dielectric strength	3.7	4.15		
Remote control			1.0%	----
Local control			1.0%	----
Interconnecting box			1.0%	
Operational inspection	3.8	4.16		
Local control		4.16.1	1.0%	----
Remote control		4.16.2	1.0%	----
Interconnecting box J-654/G	3.9	4.17	1.0%	

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:58Z
Check the source to verify that this is the current version before use.

Generator, Ringing, Hand G-42( )/PT	3.11	4.19	
Insulation			1.0% for the group
Open circuit voltage			
Power output			
Overspeed			

NOTE: Air seal test shall be performed last.

4.5.2 Group B inspection. Group B inspection shall normally be performed in inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection, including sampling, shall conform to Table IV and to the special procedures for small-sample inspection of MIL-STD-105. The reduced inspection procedure shall be R-1.

4.5.2.1 Group B sampling plans. The Group B sampling plan for the AQL's listed in Table IV shall be as follows:

AQL	Inspection level for normal and tightened inspection	Inspection level for reduced inspection
6.5%	L-7	L-5

TABLE IV - Group B inspection

Inspection	Req Para	Test Para	AQL's
Leakage	3.12.4	4.11	6.5%
Generator, Ringing, Hand G-42/PT	3.11	4.9	
Harmonic content			6.5%
Frequency			6.5%
Interchangeability	3.13	4.20	6.5%

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspection listed in Table V. Samples shall be selected in accordance with 4.5.3.1.

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:58Z
Check the source to verify that this is the current version before use.

TABLE V GROUP C INSPECTION

Inspection	Req Para	Test Para
	Subgroup I (See 4.5.3.1.1)	
Generator Ringing, Hand G-42PT	3.11	4.19
OPERATIONS		
Low Temperature	3.12.1.1	4.7
High Temperature	3.12.1.2	4.8
	Subgroup II (See 4.5.3.1.2)	
Bounce	3.12.5	4.12
	Subgroup III (See 4.5.3.1.3)	
Shock, ballistic	3.12.5	4.13
	Subgroup IV (See 4.5.3.1.4)	
Humidity	3.12.2	4.9

4.5.3.1 Sampling for Inspection of Equipment. Units selected for each Group C inspection shall be selected without regard to their quality in accordance with the following:

4.5.3.1.1 Subgroups I. For this subgroup two units from the first production lot shall be selected for the first Group C inspection. For subsequent Group C inspections, two units from each successive 100, or two per month if less than 100 are produced monthly, shall be selected.

4.5.3.1.2 Subgroup II. For this subgroup, two units from the first production lot shall be selected for the first Group C inspection. For subsequent Group C inspections, two units from each successive 200, or two every two months if less than 200 are produced every two months, shall be selected.

4.5.3.1.3 Subgroup III. For this subgroup, two units from the first production lot shall be selected for the first Group C inspection. For subsequent Group C inspections, two units every four months shall be selected.

4.5.3.1.4 Subgroup IV. For this subgroup, two units from the first production lot shall be selected for the first Group C inspection. For subsequent Group C inspections, two units every six months shall be selected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:58Z
Check the source to verify that this is the current version before use.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in an order which is satisfactory to the Government.

4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Air seal test. The equipment shall be opened and closed again in such a manner as to break and remake the seal. Immediately thereafter, the equipment as field transported shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the transit case or to the interior of the equipment inclosure when no transit case is provided. The vacuum shall then be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1 minute period the decrease in vacuum shall not exceed 0.01 pound. The gage used shall have an accuracy such that 0.01 pound can be determined readily.

4.7 Low temperature. The equipment shall be tested in accordance with Method 502.1, Procedure I, of MIL-STD-810. The storage temperature shall be -80°F and the operating temperature shall be -40°F. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.12.1.1.

4.8 High temperature. The equipment shall be tested in accordance with Method 501.1, Procedure I of MIL-STD-810. The operating temperature shall be 150°F. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.12.1.2.

4.9 Humidity. The equipment shall be subjected to the humidity test specified in Method 407.1, Procedure II, of MIL-STD-810, with test measurements during step 4, and during the 24- to 28- hour period and last 5-hour period of each cycle in step 6. Inability of the equipment to meet the requirements of 3.12.2 shall constitute failure of this test .

4.10 Elevation. The Local Control C-434( )/GRC and Remote Control C-433( )/GRC shall be tested as specified in 4.16.

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:58Z
Check the source to verify that this is the current version before use.

4.10.1 These components shall then be tested in a vacuum chamber which has been evacuated to simulate an altitude of 10,000 feet.

4.10.2 After measurements have been taken in 4.10.1, disconnect the power source and evacuate the chamber to simulate an altitude of 25,000 feet. No measurements shall be taken.

4.10.3 After one minute under the conditions specified in 4.10.2, the chamber shall be returned to a simulated altitude of 10,000 feet. The power shall be turned on and the units tested as specified in 4.16.

4.10.4 The units shall then be returned to normal atmospheric pressure and tested as specified in 4.16.

4.10.5 The units shall meet full specification performance during the tests of 4.10, 4.10.1, 4.10.3 and 4.10.4. There shall be no mechanical or electrical failures as a result of this test and all moving parts shall be in good operating condition.

4.11 Leakage test. The equipment, as prepared for field transportation, shall be tested in accordance with Method 512, Procedure I, of MIL-STD-810.

4.12 Bounce (vibration) test. The local and remote control components shall be tested for resonant frequencies below 55 cycles per second, to determine compliance with 3.12.5, as follows:

4.12.1 Vibration (resonance search).

4.12.1.1 The equipment shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. The vibration table shall provide approximately sinusoidal vibration. Internal batteries shall be in place in the equipment.

4.12.1.2 The equipment shall be vibrated successively in three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle-per-second steps. The total excursion shall be constant at 1/64 inch.

4.12.1.3 Mechanical resonance, if any, of the complete structure, of subassemblies, and of component parts shall be measured by optical or any other means, provided that the vibration of the test item is not affected by the measurement.

4.12.1.4 The equipment shall be vibrated at least 30 minutes in each of the three directions specified in 4.12.2.

Source: https://assist.dla.mil -- Downloaded: 20260-17T23:58Z
Check the source to verify that this is the current version before use.

4.12.1.5 The equipment shall have no resonance in the frequency range of 10 to 55 cycles per second that exceed twice the amplitude of the applied vibration.

4.12.2 Bounce test. The local and remote control components shall be secured to the Vehicular Adapter Plate and placed on the table of the Package Tester, both as made by the LAB Corporation, Skaneateles, New York, or equal. The plate shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 rpm ± 1 percent, for a total of 3 hours. The adapter plate shall be rotated through 90 degrees, in the same direction, at the end of each 3/4 hour period. The equipment shall meet the requirements of 3.12.5.

4.13 Shock test; ballistic. The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in MIL-S-901. The local and remote control components shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of 9 blows: one each 1-foot blow, 3-foot blow, and 5-foot blow on the back, side and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible.

4.14 Shock test, bench-handling. The chassis-and-front-panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows:

4.14.1 Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to horizontal. Repeat, using the other practicable edges of the same horizontal face as axes, for a total of four drops.

4.14.2 Repeat 4.14.1 with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly could be placed practicably during servicing. The requirement of 3.12.5 shall be met.

4.15 Dielectric test. The dielectric strength of all circuits shall be tested to determine compliance with 3.7. The test voltage shall be d-c or peak a-c of the same value and shall be applied for at least 10 seconds between wiring and adjacent nonconductors and ground.

4.16 Operational test (See 3.8).

4.16.1 Local control C-434( )/GRC. The Local Control C-434( )/GRC shall be connected to a Remote Control C-433( )/GRC by means of two artificial lines, one representing two miles and the other one quarter mile of wet Wire WD-1/TT. A Handset H-33( )/PT shall be connected into each control. Each unit shall contain batteries for operation. The Local control shall be tested as follows:

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:58Z
Check the source to verify that this is the currentversion before use.

Switch Positions			Operation	Required Result
C-433	C-434			
"Selector" Switch	"Remote" Switch	"Local" Switch		
1. Tel	Tel Only	Tel	Operate Generator on C-433	Ringer or Light shall operate on C-434. Internal switch shall be operated to obtain operation of Light & Ringer.
			Operate Generator on C-434	Ringer or Light shall operate on C-433
			Operate Handsets at C-433 & C-434	Full duplex telephone communication shall be established.
2 Left	Set 1 & 2	Tel	Press switch of handset @ C-433	Radio Set 1 mic. control line shall be grounded
				Audio from C-433 shall appear @ the mic. Audio lines to Radio Sets 1 and 2.
3 Center	Set 1 & 2	Tel	Press switch of handset @ C-433	Radio Set 2 mic. control line shall be grounded.
				Audio from C-433 shall appear @ the mic. Audio lines to Radio Sets 1 and 2.
4 Left	Set 1	Tel	Press and then release the switch of handset @ C-433	Remote battery line shall be grounded.

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:58Z
Check the source to verify that this is the current version before use.

			Press & hold switch of handset @ C-433	Radio Set 1 mic. control line shall be grounded. Audio from C-433 shall appear @ the mic audio line to Radio Set 1.
5 Center	Set 1	Tel	Press and then release switch of handset @ C-433	Remote battery line shall be ungrounded
			Press switch of handset @ C-433	Radio Set 1 mic. control line shall be ungrounded.
6 Left	Set 2	Tel	Press & then release the switch of handset @ C-433	Remote battery line shall be grounded
			Press and hold switch of handset @ C-433	Radio Set 2 mic. control line shall be grounded. Audio from C-433 shall appear @ the mic audio line to Radio Set 2.
7 Center	Set 2	Tel	Press and then release switch of handset @ C-433	Remote battery line shall be ungrounded.
				Radio Set 2 mic. control line shall be ungrounded.
8 Left or Center	Set 1 &2 or Set 1 or Set 2	Set 1	Press switch of handset @ C-434	Radio Set 1 mic. control line shall be grounded. Audio from C-434 shall appear @ Set 1 audio line.
9 Left or Center	Set 1 &2 or Set 1 or Set 2	Set 2	Press switch of handset @ C-434	Radio Set 2 mic. control line shall be grounded. Audio from C-434 shall appear @ Set 2 audio line.

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:58Z
Check the source to verify that this is the currentversion before use.

4.16.2 Remote Control C-433( )/GRC. Using the test set-up of 4.16.1, repeat test no. 1, 2 and 3 of 4.16.1 with the following change: In test no. 1, operate the internal switch of C-433 to obtain operation of both the light and ringer from the generator of C-434. Operation of the internal switch of C-434 may be omitted in this test.

4.17 Interconnection box J-654( )/GRC test. Test the box for proper circuit continuity (See 3.9).

4.18 Bag CW 189/GR test. The units shall be placed in the bag to determine compliance with 3.10.

4.19 Generator, Ringing, Hand G-42( )/PT tests. The generator shall be tested in accordance with the following paragraphs in MIL-G-11994 and shall meet the requirements of 3.11:

4.5.1.2 Insulation test
4.5.1.3 Open circuit voltage test
4.5.1.4 Power output test
4.5.1.5 Overspeed test
4.5.1.6 Harmonic content test
4.5.2.2 Operations
4.5.2.2 Frequency
4.5.2.3 Starting torque
4.5.2.4 Switch operation
4.5.2.5 Switch click elimination

4.20 Inspection for dimensional interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.13. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, slides and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:58Z
Check the source to verify that this is the current version before use.

4.21 Bounce preconditioning. The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operating position on the table of the Package Tester as made by the LAB Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick-strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to procurement inspection.

4.22 Visual and mechanical inspection. The equipment shall be examined for the defects listed in MIL-STD-252.

4.23 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to MIL-P-12292. Rough handling test is required only when specified in the contract or order.

5. PACKAGING

5.1 Preservation, packaging, packing, marking and test requirements shall be in accordance with MIL-P-12292.

6. NOTES

6.1 Intended use. The equipment provides local and remote control facilities for the standardized series radio sets. It also provides telephone communication between the local and remote sites.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Type required.

c. Level of packaging and level of packing required for shipment. (Level A, Level B or Level C.)

d. Place of final inspection.

e. Technical literature required.

f. Quantity of tools and running spare parts required.

6.3 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example AN/GRA-6W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:58Z
Check the source to verify that this is the current version before use.

6.4 Location of operational inspection and air seal test. It is desirable that the operational inspection (4.16) and the air seal test (4.6) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.5 Color. The color chip furnished by the procuring agency (3.4) will match color Chip No. X-24087, and may be obtained upon request to Commander, US Army Electronics Command, Procurement and Production Directorate, Fort Monmouth, New Jersey 07703.

6.6 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum use of the contractor's quality control system and the quality history of the product.

Custodian
Army-EL

Preparing activity:
Army-EL

Project Number 5820-A746

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:58Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-R-13028B(CR)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

HQ. USA CORADCOM
ATTN: DRDCO-CM-DM
Fort Monmouth, NJ 07703

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:58Z
Check the source to verify that this is the current version before use.